Why Some Brains Find Math So Hard

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For some children, math is much harder than for their classmates. Even with practice, they keep having trouble with problems others can solve. A new Stanford study may explain why.

The study looked at children with developmental **dyscalculia**, which makes math hard. The team studied 68 children ages 8 to 10.

Solving a math problem takes more than knowing numbers. A child must also pick the best way to solve it, and knowing when to **switch** matters more as problems get harder.

Children with **dyscalculia** were not as good at using different ways to solve problems. They took longer to **switch**, and they were slower to change when a problem got harder. Other children got better. They did not.

Brain scans gave another clue. One part of the brain worked differently in children with **dyscalculia**, and the scans could guess how well a child counted and **switched**.

The study shows that math trouble may come from skills like paying **attention**, **memory**, and changing your thinking. Learning how children pick ways to solve problems may help these children sooner.

Word Treasure

dyscalculia -- A condition that makes math hard for a long time.
brain scans -- Pictures of the inside of the brain made by a machine.
switch -- To change from one way of doing something to another.
attention -- Focusing your mind on one thing.
memory -- The ability to remember things.

Background you need

Math is the study of numbers and how to solve problems with them.

The brain is the organ in your head that controls thinking and learning.

Some children find math hard, even when they work at it, and scientists study why.

Break it down

WHO: Children with developmental dyscalculia, studied by Stanford researchers
WHAT: A study of why some children find math so hard
WHERE: At Stanford University
WHY: To learn how children pick ways to solve problems, so they can be helped sooner

Why it matters

If math is hard for you, this study shows it is not your fault, and teachers may soon know better ways to help you.

Test what you remember

1. What condition did the study look at?

- (A) A cold
- (B) Poor eyesight
- (C) Developmental dyscalculia
- (D) A broken arm

2. How many children did the team study?

- (A) 68
- (B) 540
- (C) 10
- (D) 8

3. How old were the children?

- (A) 12 to 14
- (B) 18 to 20
- (C) 5 to 6
- (D) 8 to 10

4. What were children with dyscalculia less good at?

- (A) Counting to ten
- (B) Drawing pictures
- (C) Reading books
- (D) Using different ways to solve problems

5. What did the brain scans show?

- (A) One part of the brain worked differently
- (B) The brain was the same in everyone
- (C) The brain had no activity
- (D) The brain was smaller

6. What skills may math trouble come from?

- (A) Cooking and cleaning
- (B) Paying attention, memory, and changing your thinking
- (C) Singing and dancing
- (D) Running and jumping

Answer key (try the quiz first!): 1: C 2: A 3: D 4: D 5: A 6: B

Questions to think about

MY THOUGHTS (at least 20 words):